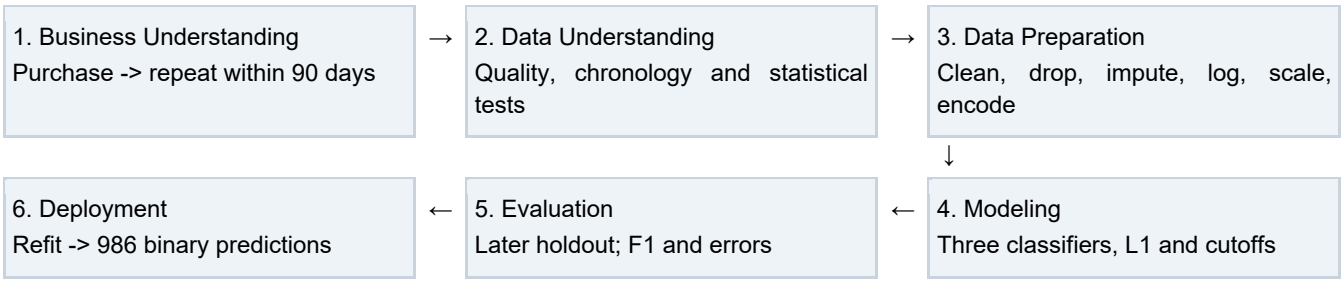
Predicting Repeat Purchases

Machine Learning 2026/27 | Homework | Group 06
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Code: https://github.com/MiguelSantos00/Predicting-Repeat-Purchases-ML



1. Problem Scope and Data Foundations

Our objective is to predict whether a wholesale client of Everything & Then Some, Ltd. will complete a subsequent purchase within 90 days following a given order event (repeat_purchase_90d). Because each row represents a distinct transactional occasion, Binary F1 is the official Kaggle metric. individual clients can recur over time. Model performance is judged strictly by binary F1-score on the returning class. The benchmark dataset supplies 6,534 historical training rows, 986 out-of-time test instances, and a complementary table with 1,143 client profiles. Merging clientes.csv via a left join on customer_id achieved a 100% match across all transaction records.

2. Preparation and rationale

Before training, we removed 10 rows that were exact duplicates apart from their ID, and 55 records with no date and no purchase information, leaving 6,469 training purchases. All 986 test rows were kept in their original order. We dropped the ID and customer_id columns, the empty customer_type column, company name and address, and the most detailed location fields (street and postal code), which almost identify individual customers. The remaining geographic codes are treated as categories, not distances.

Negative gaps and values, which are impossible, and the repeated value 1,000,000,000, which looks like a placeholder, were set to missing. Where the historical mean was missing or did not match total spend divided by purchase count, we recalculated it from that formula; when the count is zero, it stays undefined. Historical maxima above total spend, and annual spend below 180-day spend or above lifetime spend, were set to missing. Large amounts consistent with the rest of the record were kept, as they reflect important wholesale clients. Instead of dropping variables blindly, we tested each one against the target on development data: ANOVA, Mann-Whitney U and mutual information for numeric inputs, and chi-square with Cramér's V for categorical ones. random_noise (ANOVA $p = 0.163$, Mann-Whitney $p = 0.300$, mutual information 0.005) and Total Portes ($p = 0.401$ and 0.255) showed no relationship with repeat purchases and were dropped. High Cramér's V for fields like city mostly comes from small categories that nearly identify single customers; encoding groups categories with fewer than 20 training purchases.

We also excluded the purchase year. Test purchases are from 2025 and 2026, so a pattern such as more repeat purchases in 2024 would not help predict them. Month and weekday were kept as candidate inputs, although the dates are anonymised. Remaining numeric gaps were filled with training-fold medians, with a flag marking each filled value, and missing text became "Missing". Monetary variables were log-transformed (log1p) before standard scaling.

3. Temporal assessment

Because the test purchases come after the training period, we split the data chronologically instead of randomly, to avoid temporal leakage. We sorted all dated purchases and kept the last 20% (1,298 purchases from 17 June 2023 onward) as a later holdout. Development uses the 5,024 purchases up to 18 March 2023. The 147 purchases in between form a 90-day gap, so every development outcome is already known before the holdout starts. Purchases on the same day stay together.

Within development, we used three expanding temporal folds, each with its own 90-day gap. Imputation, scaling, encoding and L1 selection were fitted only on the training part of each fold and then applied to its validation part. The model, feature set and decision cutoff were chosen by mean F1 across these folds, before the final model was evaluated once on the holdout.

4. Selection and classifier comparison

For feature selection we used L1-penalised logistic regression (C = 0.1), keeping variables with an absolute coefficient above 0.000001. It was fitted within each training fold, so it never saw the validation data. Each classifier was tested with all inputs and with the L1 subset: logistic regression (C = 1), a decision tree (depth 5, minimum leaf 20) and a random forest (200 trees, minimum leaf 5), against a majority-class baseline. Seeds were fixed and hyperparameters were set in advance, not tuned.

By default a classifier predicts a repeat purchase above a probability of 0.5, which is often not the best cutoff for F1. We therefore tested cutoffs from 0.20 to 0.60 for every model on the same three development folds:

Candidate / features	F1 at 0.5	Best cutoff	F1 at best cutoff	Fold SD
Baseline / All	0.7044	any	0.7044	0.0243
Logistic regression / All	0.6778	0.30	0.7186	0.0210
Logistic regression / L1	0.7294	0.40	0.7431	0.0076
Decision tree / All	0.7223	0.40	0.7283	0.0073
Decision tree / L1	0.7232	0.40	0.7271	0.0076
Random forest / All	0.7376	0.40	0.7457	0.0125
Random forest / L1	0.7388	0.40	0.7451	0.0132

Every classifier improved when the cutoff was lowered from 0.50. The two random forests performed almost identically: a difference of only 0.0006, far smaller than their fold standard deviations (about 0.013). We therefore chose the simpler option, random forest + L1 with a 0.40 cutoff, which reduces the full-development inputs from 273 to 52. The fold standard deviation only shows how much the scores vary between folds; it is not a confidence interval.

5. Fixed later holdout

Solution	F1	Precision	Recall	Accuracy
Baseline	0.7525	0.6032	1.0000	0.6032
Forest + L1, cutoff 0.5	0.7818	0.7979	0.7663	0.7419
Forest + L1, cutoff 0.40	0.7915	0.7209	0.8774	0.7211

The final model improves F1 by 3.9 points over the baseline. About 1 point of that comes from lowering the cutoff from 0.5 to 0.40, which raises F1 from 0.7818 to 0.7915. The confusion counts are TN = 249, FP = 266, FN = 96 and TP = 687. The lower cutoff gives up some accuracy in exchange for higher recall: the model now finds 87.7% of the customers who really bought again. The brief gives no contact costs or margins, so we do not estimate the financial effect.

F1 is 0.8008 for customers already seen in development and 0.7000 for customers who were not, although 101 of those 145 purchases still had earlier history. First purchases are the hardest case. Of the 44 in the holdout, 12 were followed by another purchase within 90 days. The model finds 5 of them at the 0.40 cutoff (F1 0.4348), against none at 0.5. With no purchase history, their predicted probabilities rarely reach 0.5. None of our choices was based on holdout results: variables were removed using development-data tests, and the cutoff was chosen on the development folds.

6. Refit, export and limits

For the Kaggle submission, we retrained the same pipeline on all 6,469 dated training purchases, now including the holdout and the 147 purchases from the former gap. The test set starts 98 days after the last training purchase, so all training outcomes were already known. Preparation was refitted on this full data, producing 299 encoded columns. L1 selection kept 56 of them before the random forest was refitted and the 0.40 cutoff applied.

The submission file, ML_Group_06_Version03.csv, contains one prediction for each of the 986 test rows, without an extra index: 767 repeat purchases and 219 non-repeats. These counts describe the predictions, not their accuracy on Kaggle. The earlier versions are kept for comparison: Version 01 still used random_noise, Total Portes and the purchase year (holdout F1 0.7778), and Version 02 removed them but kept the default 0.5 cutoff (holdout F1 0.7818). The six numbered notebooks on the repository's homework branch reproduce the file from the raw data.